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From

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To

The Academic Director
E.G.S. Pillay Engineering College
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Subject: Request for Credit Points for Published Q1 Research Article.

(Publishing in Tier-1 Journals (Q1/Q2, High Impact, Highly Cited))

Respected Sir,

I wish to inform you that I have published a research article entitled “Sustainable LDPE composites reinforced with Syzygium cumini seed biofillers: Mechanical, thermal, and morphological performance assessment” in the Journal of [Next Materials](#), Elsevier publisher (Online ISSN: 2949-8228), which is in the Tier-1 Journals Q1 category. In this regard, I requested that you consider my research article for faculty credit points, and I have attached my research article details for reference.

Thank you.

Yours sincerely,

N. Sridhar

Enclosure:

1. Research article
2. Proof for Tier-1 Journals in the Q1 category
3. proof for indexing in Scopus and Web of Science



Research article

Sustainable LDPE composites reinforced with Syzygium cumini seed biofillers: Mechanical, thermal, and morphological performance assessment

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ABSTRACT

Environmentally sustainable polymer composites utilizing agro-waste biofillers provide a responsible method for improving thermoplastic performance. This work involved the fabrication of low-density polyethylene composites supplemented with Syzygium cumini seed biofillers using injection molding, utilizing filler loadings of 0, 10, 20, 30, and 40 wt%. The mechanical, physical, thermal, and morphological qualities were extensively assessed. The findings indicated that mechanical performance enhanced with rising filler content, peaking at 20 wt% Syzygium cumini, where the composite had a tensile strength of 6.58 MPa, an elastic modulus of 31.4 MPa, and a Shore D hardness of 42. The density grew significantly with the addition of filler, showing the successful assimilation of the biofiller. Dynamic mechanical investigation indicated superior stiffness retention at 20 wt% Syzygium cumini, whereas 10 wt% Syzygium cumini exhibited higher damping characteristics. Fourier-transform infrared spectroscopy and Energy-dispersive X-ray spectroscopy investigations validated the existence of lignocellulosic functional groups and bio-derived elemental composition, whereas scanning electron microscopy observations revealed homogeneous filler dispersion at optimal loading levels. Syzygium cumini seed particles are an economical and sustainable reinforcing filler for low-density polyethylene, with 20 wt% determined as the ideal reinforcement concentration.

1. Introduction

Polymer-based composites have become appealing substitutes for metallic materials due to their low density, corrosion resistance, cost-effectiveness, and ease of processing, rendering them appropriate for applications in packaging, automotive, and aerospace structures [1,2]. Low-density polyethylene (LDPE) is one of the most extensively utilized commercial polymers, especially in food packaging, agricultural films, containers, electronic products, shrink-wrapping, and other household items [3]. LDPE features an extensively branched molecular architecture that enhances its low density, remarkable flexibility, superior impact

resistance at low temperatures, and robust resistance to chemicals and creep deformation. Its versatility across various manufacturing processes, coupled with a recycling potential of approximately 95%, significantly amplifies its commercial significance [4–7]. Notwithstanding its benefits, the extensive use of LDPE presents a considerable waste management dilemma. The Central Pollution Control Board (CPCB) indicated that of the 1.28 million tons of plastic garbage produced each year in 60 prominent Indian cities, merely 0.77 million tons are recycled, whereas the residual 0.51 million tons are either landfilled or discarded as waste. This underscores the pressing necessity for environmentally viable ways to augment and partially substitute

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